

Supporting information

Table S1. Functional hypotheses of morphological and thermal traits.

Trait & Measurement	Hypothesized link to habitat conversion
Body size WL Weber's Length Diagonal length of mesosoma length	Responds to the habitat complexity, i.e. the size-grain hypothesis (Kaspari & Weiser, 1999); food recourse type and its richness (Silva & Brandao, 2010); heat loss rate about the body size and ambient temperature (Bergmann, 1848).
REW Relative Eye Width Maximum width for the eyes corrected by body size (Eye width / Weber's length)	Responds to the habitat complexity and allows to detect their predator and prey (Silva & Brandao, 2010).
REP Relative Eye Position (Head width – Interocular distance) / Head length	Responds to the visual ability and diet preference, and species with larger REP are determined as visual predators (Schofield <i>et al.</i> , 2016; Sosiak & Barden, 2020).
RFL Relative hind leg Femur Length Maximum length for the femur of high legs corrected by body size (Eye width / Weber's length)	Determines mobility in terms of foraging for recourses, escaping from predators and heat shock, aggregating for ants (Grevé <i>et al.</i> , 2019).
RSL Relative Scape Length Maximum length for the first antenna corrected by body size (Scape length / Weber's length)	Determines the searching and navigating abilities in responding different niches and adapting different habitats (Silva & Brandao, 2010).
RMS Relative Mandible Size Mandible length / head width	Responds to diet preference of ant predator and environment complexity where should have more species with larger RMS in complex forest habitat due to the foraging specialization hypothesis (Guilherme <i>et al.</i> , 2019).
CT_{max}	Responds to the ambient temperature and determines species performance in function of

Critical Thermal maximum	temperature (Lutterschmidt & Hutchison, 1997).
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Table S2. Functional traits of species. Abbreviation of traits: CTmax, Critical Thermal maximum; WL, Weber's length; RMS, relative mandible size; REP, relative eye position; REW, relative eye width corrected by body size; RSL, relative scape length corrected by body size; RHF, relative hind femur length corrected by body size.

Species	Habitat	N_ctm ax	CTm ax	W L	R MS	R EP	RE W	R SL	R HF
<i>Aenictus</i> sp. cf <i>parahuonicus</i>	forest	10	43.60	1. 25	0.5 2	N A	NA	0.5 2	0.7 7
<i>Aenictus parahuonicus</i>	forest	10	44.33	1. 21	0.4 9	N A	NA	0.4 9	0.9 0
<i>Anoplolepis gracilipes</i>	forest	15	48.60	1. 71	0.5 6	0.1 8	0.1 4	1.0 3	1.4 4
<i>Carebara</i> sp. 1	forest	10	45.90	0. 39	0.4 4	0.0 8	0.0 6	0.5 9	0.6 0
<i>Carebara</i> sp. 2	forest	10	47.90	0. 95	0.6 2	0.0 7	0.1 4	0.7 6	0.9 9
<i>Crematogaster</i> sp. 1	forest	10	51.70	0. 49	0.4 8	0.1 4	0.2 1	0.6 8	2.4 1
<i>Ectomomyrmex astutus</i>	forest	10	47.00	4. 14	0.7 2	0.2 1	0.1 1	0.5 4	0.6 3
<i>Euprenolepis</i> sp. cf <i>procera</i>	forest	23	45.70	1. 48	0.5 6	0.3 7	0.2 0	1.1 0	1.1 2
<i>Leptogenys</i> sp. 1	forest	10	43.80	2. 47	0.7 3	0.2 2	0.1 4	0.7 0	0.8 3
<i>Lophomyrmex bedoti</i>	forest	10	44.20	0. 96	0.4 8	0.1 1	0.1 5	0.6 5	0.7 0
<i>Monomorium</i> sp. 1	forest	10	47.30	0. 42	0.5 5	0.0 9	0.1 4	0.5 9	0.6 0
<i>Nylanderia</i> sp. 1	forest	20	46.20	0. 56	0.4 8	0.2 5	0.1 8	0.9 1	0.8 2
<i>Odontomachus</i> sp. 1	forest	18	43.22	3. 08	0.8 1	0.1 9	0.1 2	0.8 4	0.8 8
<i>Odontoponera denticulata</i>	forest	18	46.50	3. 55	0.5 7	0.2 3	0.1 2	0.5 3	0.6 8
<i>Pheidole</i> sp. 1	forest	45	45.58	1. 56	0.7 1	0.1 6	0.1 5	1.1 1	1.3 9
<i>Pheidole</i> sp. 2	forest	12	41.83	0. 63	0.5 6	0.1 6	0.1 4	0.7 2	0.7 6
<i>Pheidole</i> sp. 3	forest	10	45.30	0. 47	0.6 2	0.1 6	0.2 0	0.7 4	0.7 9
<i>Pheidole</i> sp. 4	forest	10	45.20	0. 94	0.6 6	0.2 0	0.1 8	0.9 1	0.9 4
<i>Tetramorium</i> sp. 1 <i>obesum</i> group	forest	15	46.40	0. 74	0.5 7	0.1 4	0.1 9	0.5 3	0.6 1
<i>Anochetus graeffei</i>	plantati on	14	44.71	1. 30	0.6 0	0.2 4	0.1 2	0.5 6	0.5 9
<i>Anoplolepis gracilipes</i>	plantati on	19	49.32	1. 71	0.5 6	0.1 8	0.1 4	1.0 3	1.4 4
<i>Camponotus</i> sp. 1 brown	plantati on	15	48.40	2. 65	0.5 2	0.2 6	0.1 9	0.9 7	1.0 0
<i>Camponotus</i> sp. 2 dark	plantati on	3	50.00	2. 62	0.5 2	0.2 4	0.1 6	0.8 5	1.0 0
<i>Diacamma</i> sp. <i>rugosum</i> complex	plantati on	7	51.29	3. 42	0.7 2	0.2 5	0.1 8	0.7 5	0.8 4
<i>Euponera sharpi</i>	plantati on	13	44.69	1. 55	0.6 4	0.1 4	0.0 5	0.4 6	0.5 7
<i>Stictoponera</i> sp. cf <i>binghamii</i>	plantati	14	46.07	1.	0.5	0.1	0.1	0.5	0.6

	on			55	8	0	3	5	3
<i>Stictoponera</i> sp.	plantati	10	46.00	1.	0.5	0.1	0.1	0.5	0.6
	on			54	5	9	5	4	9
<i>Lepisiota</i> sp. cf <i>rothneyi</i>	plantati	12	55.33	0.	0.5	0.3	0.2	0.8	0.8
	on			83	1	4	3	0	5
<i>Odontoponera denticulata</i>	plantati	10	47.60	3.	0.5	0.2	0.1	0.5	0.6
	on			55	7	3	2	3	8
<i>Tetramorium</i> sp. 2 <i>obesum</i>	plantati	10	44.90	0.	0.4	0.1	0.2	0.6	0.7
group	on			70	7	7	1	3	6

Table S3. Results of regression models using land-use as the predictor at multiple levels. Ordinary least squares (OLS) models.

	Estimate	SE	DF	<i>p</i> -value	R2
WL	0.1976	0.1098	28	0.0826	0.1037
RMS	-0.0104	0.0250	28	0.6790	0.0062
REP	0.1043	0.0676	26	0.1350	0.0839
RSL	-0.0217	0.0438	28	0.6240	0.0087
RHF	-0.0433	0.0542	28	0.4310	0.0223
REW	-0.0046	0.0578	26	0.9370	0.0002

Table S4. Results from principal component analysis (PCA). The eigenvalues, proportion of inertia explained by each of the first three components (PC1, 2, 3) and trait loadings are presented for three functional trait space (both land-use types, forest and rubber plantation only). Abbreviations as in Table S1.

	PC1	PC2	PC3
<i>Both habitats</i>			
Eigenvalue	1.9778	1.3736	0.8039
Proportion Explained	39.5552	27.4728	16.0783
Cumulative Proportion	39.5552	67.0280	83.1063
CTmax	0.3155	0.7348	-0.4224
WL	0.7707	-0.3977	-0.3583
RMS	0.6102	-0.6111	0.2268
REP	0.7981	0.2342	-0.0945
RSL	0.5242	0.4973	0.6608
<i>Forest</i>			
Eigenvalue	2.2262	1.1482	0.8151
Proportion Explained	44.5230	22.9638	16.3022
Cumulative Proportion	44.5230	67.4868	83.7891
CTmax	-0.3265	0.6586	0.6736
WL	0.7954	-0.2921	0.4381
RMS	0.8132	-0.1420	0.1179
REP	0.7699	0.2729	-0.0458
RSL	0.4826	0.7311	-0.3918
<i>Rubber plantation</i>			
Eigenvalue	2.2800	1.5252	0.5722
Proportion Explained	45.6003	30.5043	11.4436
Cumulative Proportion	45.6003	76.1046	87.5482
CTmax	0.8896	0.0788	0.1766
WL	0.2153	0.8550	-0.3559
RMS	-0.2085	0.8811	0.2688
REP	0.8230	0.0228	0.4054
RSL	0.8494	-0.1050	-0.4216

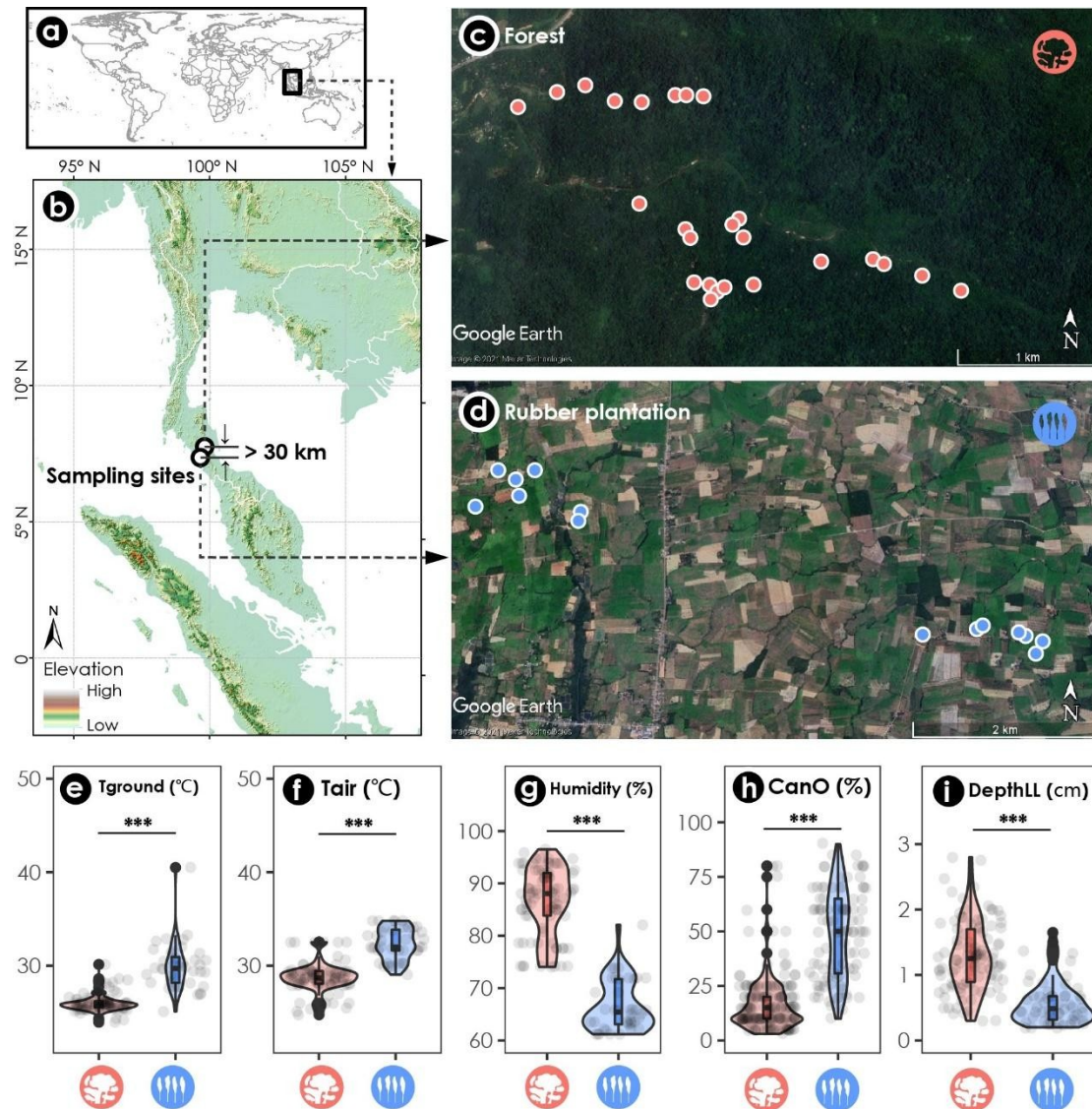


Figure S1. (a) - (d) Sampling area in southern Thailand. Environmental factors measured in the field: (e) air temperature (T_{air}), (f) ground temperature (T_{ground}), (g) humidity, (h) canopy openness (CanO) and (i) depth of leaf litter (DepthLL). Modified t-test and spatial autocorrelations between (j) air and ground temperature, (k) canopy openness and ground temperature and (l) air temperature and humidity. Red and blue color in figures represent forest and rubber plantation respectively, the same in the following figures. Asterisks indicate statistical significance (***) : $p < 0.001$, ^{ns} not significant).

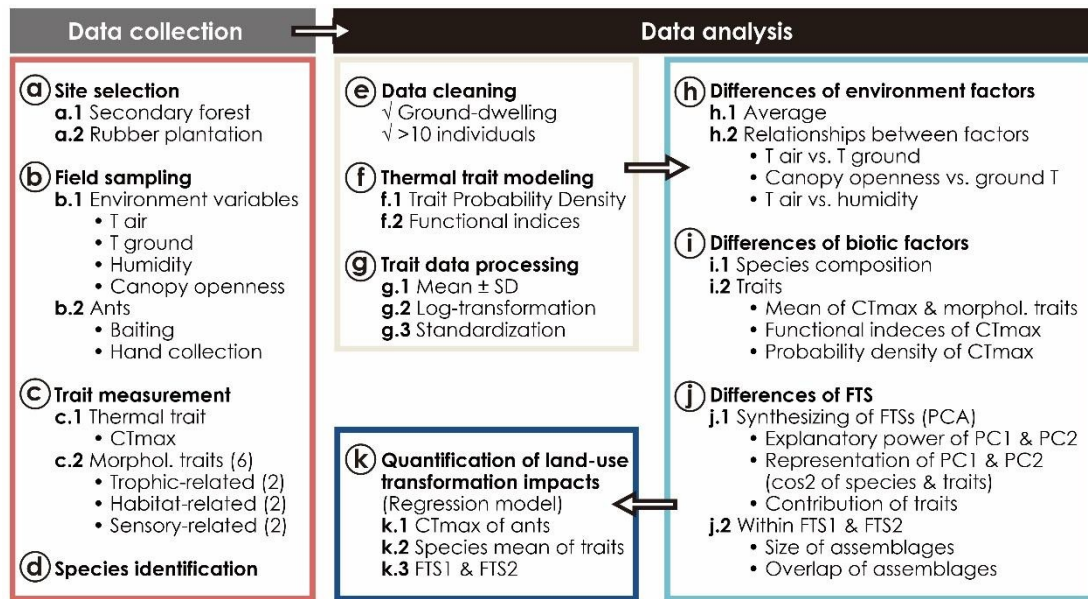


Figure S2. Analytical flow diagram of this study for the trait-based approaches of ant assemblages between rubber plantations and forests in Thailand. Abbreviation: T air, air temperature; T ground, ground temperature; Morphol. trait, morphological traits; SD, standard deviation; CT_{max}, Critical Thermal maximum; FTS, functional trait space; PCA, principal component analysis; PC, principle component.

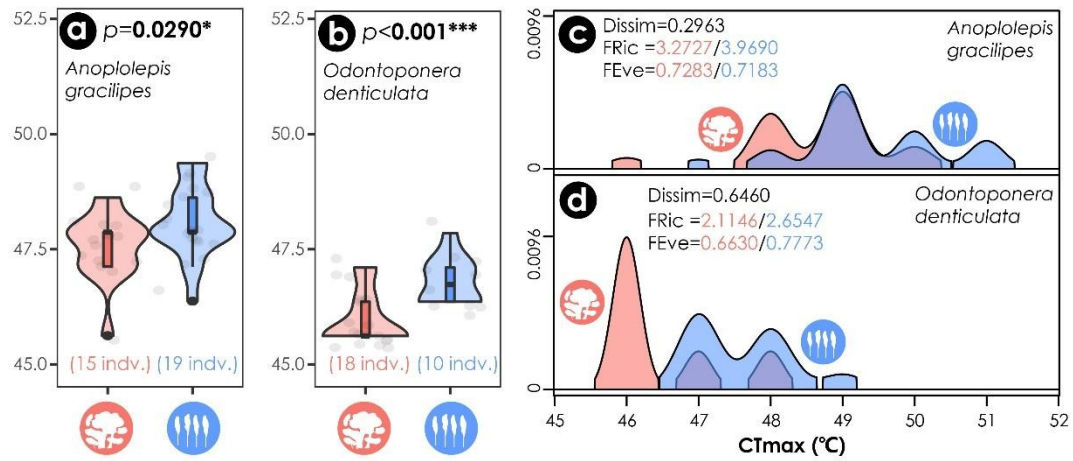


Figure S3. Conspecific Critical Thermal maximum (CT_{max}) in different habitats for (a) *Anoplolepis gracilipes* and (b) *Odontoponera denticulata*. Trait Probability Density (TPD) of CT_{max} of (c) *A. gracilipes* and (d) *O. denticulata*. Purple color represents the overlapped TPD of populations in forests and in rubber plantations. The functional dissimilarity (Dissim), Functional Richness (FRic) and Functional Evenness (FEve) in TPD of CT_{max} of two habitats are also showed on the graphs. Asterisks indicate statistical significance ($^{***} p < 0.001$, $^* p < 0.05$) from nonparametric Mann-Whitney U test.

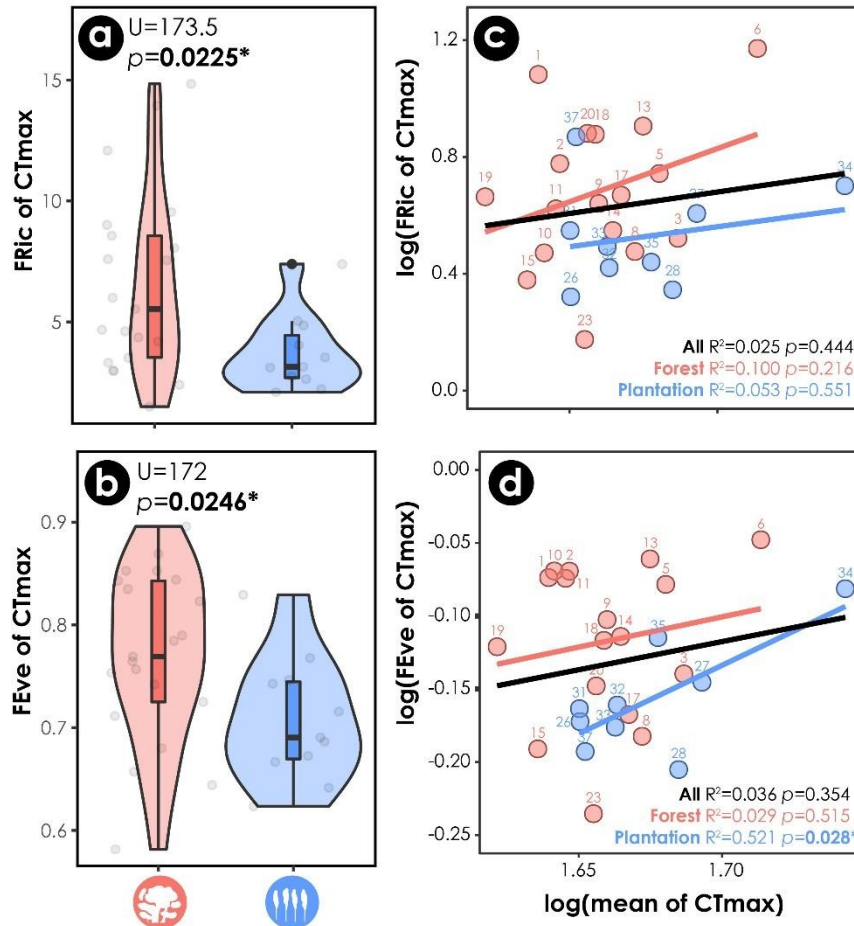


Figure S4. Intraspecific variation of Critical Thermal maximum (CT_{max}) of ant assemblages in forest and rubber plantation. (a) Functional Richness (FRic) and (b) Functional Evenness (FEve) based on Trait Probability Density (TPD) of CT_{max} in species level and (c-d) their relationship with average value of CT_{max}. Values are log-transformed to reduce heteroscedasticity in correlation analysis. R^2 and p -value are shown, asterisk indicates statistical significance (* $p < 0.05$; ^{ns} not significant).

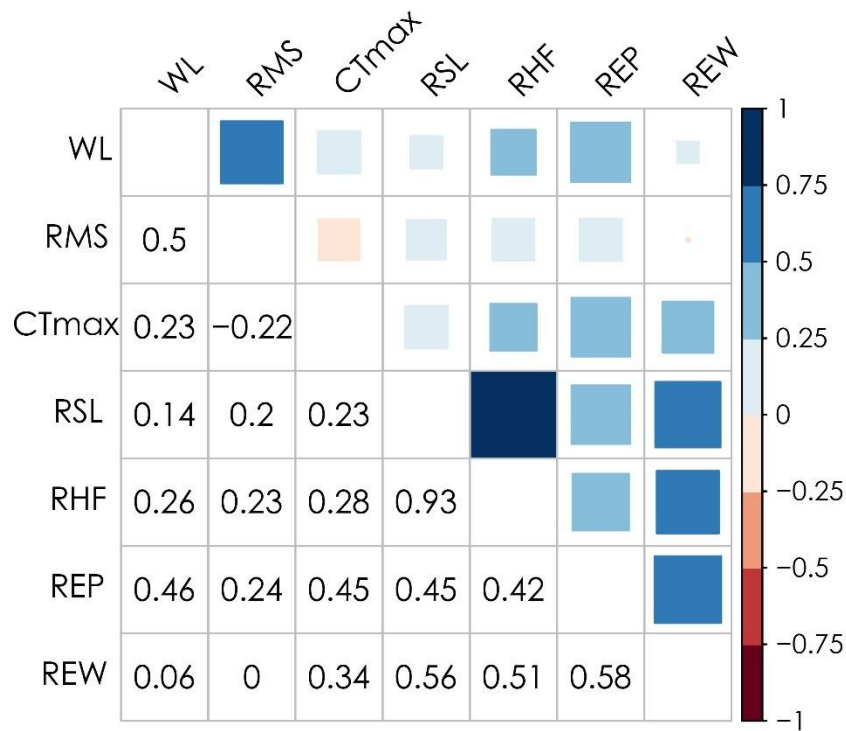


Figure S5. Pairwise correlation coefficients between all functional traits. Correlation strength is shown by colours and sizes of squares at the top right; coefficient values are presented at the bottom left. WL, Weber's length; RMS, relative mandible size; CTmax, Critical Thermal maximum; RSL, relative scape length corrected by body size; RHF, relative hind femur length corrected by body size; REP, relative eye position; REW, relative eye width corrected by body size. RHF and REW are highly correlated with other traits and thus excluded in the principal component analyses.

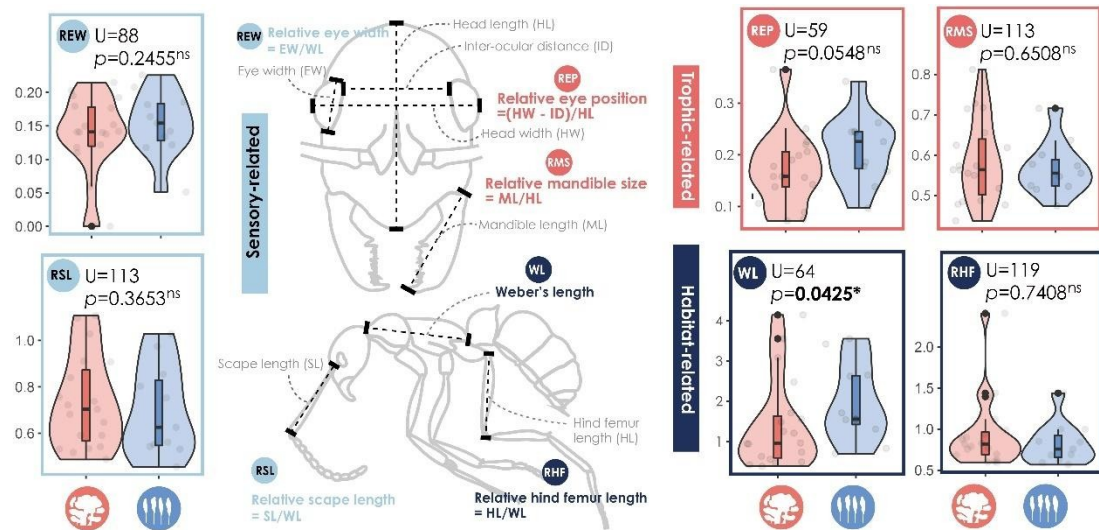


Figure S6. Six morphological traits of ant assemblages from forest and rubber plantation. Asterisks indicate statistical significance (* $p < 0.05$, ^{ns} not significant).

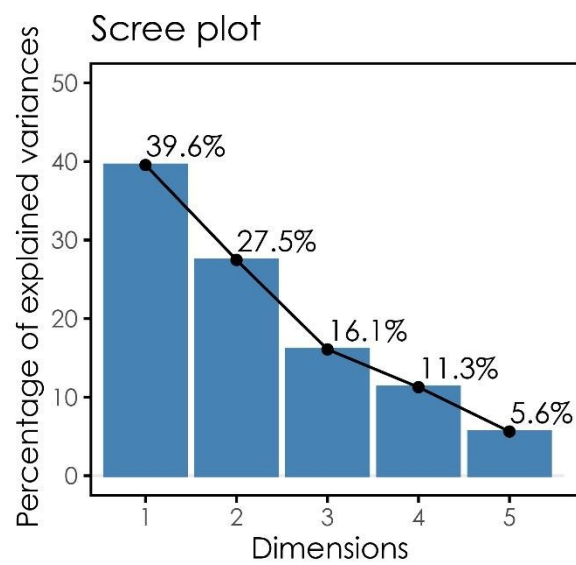


Figure S7. Scree plot for the principal component analysis (FTS1). The first three dimensions represent over 80% of variances.

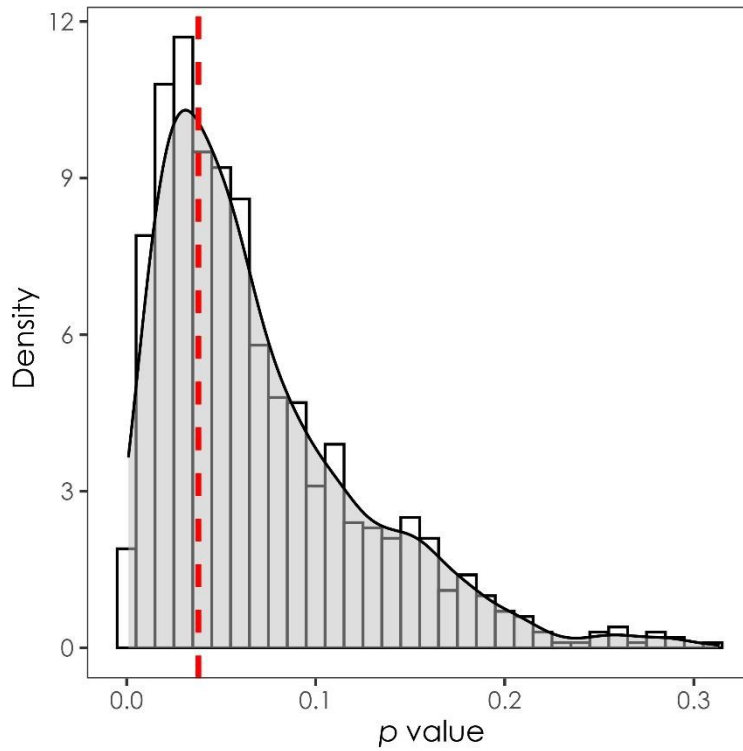


Figure S8. Density distribution of p -value from permutational multivariate analysis of variance between two ant assemblages on the first and third principal components (i.e. PC1 and PC3) by randomly rarefied forest species assemblages for 1000 times. Red dashed line indicates the p -value of observed variance which is not under-estimated due to the variation of species richness comparing to the null variance controlled for species richness ($p = 0.29$).

References

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